

New battery technologies demand new electrolyte concepts?!

																		He 2 Helium																			
H 1 Hydrogen		He 2 Helium		Li 3 Lithium		Be 4 Beryllium				B 5 Boron		C 6 Carbon		N 7 Nitrogen		O 8 Oxygen		F 9 Fluorine		Ne 10 Neon																	
Na 11 Sodium		Mg 12 Magnesium				Al 13 Aluminum		Si 14 Silicon		P 15 Phosphorus		S 16 Sulfur		Cl 17 Chlorine		Ar 18 Argon																					
K 19 Potassium		Ca 20 Calcium		Sc 21 Scandium		Ti 22 Titanium		V 23 Vanadium		Cr 24 Chromium		Mn 25 Manganese		Fe 26 Iron		Co 27 Cobalt		Ni 28 Nickel		Cu 29 Copper		Zn 30 Zinc		Ga 31 Gallium		Ge 32 Germanium		As 33 Arsenic		Se 34 Selenium		Br 35 Bromine		Kr 36 Krypton			
Rb 37 Rubidium		Sr 38 Strontium		Y 39 Yttrium		Zr 40 Zirconium		Nb 41 Niobium		Mo 42 Molybdenum		Tc 43 Technetium		Ru 44 Ruthenium		Rh 45 Rhodium		Pd 46 Palladium		Ag 47 Silver		Cd 48 Cadmium		In 49 Indium		Sn 50 Tin		Sb 51 Antimony		Te 52 Tellurium		I 53 Iodine		Xe 54 Xenon			
Cs 55 Cesium		Ba 56 Barium		* 57-70 Lanthanide series		Lu 71 Lutetium		Hf 72 Hafnium		Ta 73 Tantalum		W 74 Tungsten		Re 75 Rhenium		Os 76 Osmium		Ir 77 Iridium		Pt 78 Platinum		Au 79 Gold		Hg 80 Mercury		Tl 81 Thallium		Pb 82 Lead		Bi 83 Bismuth		Po 84 Polonium		At 85 Astatine		Rn 86 Radon	
Fr 87 Francium		Ra 88 Radium		** 89-102 Actinide series		Lr 103 Lawrencium		Rf 104 Rutherfordium		Db 105 Dubnium		Sg 106 Seaborgium		Bh 107 Bohrium		Hs 108 Hassium		Mt 109 Meitnerium		Uun 110 Ununennium		Uuu 111 Ununtrium		Uub 112 Ununbium				Uuq 114 Ununquadium									

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Layout & Learning Outcomes

Next Generation Batteries (NGBs) – a crash-course

- Why? Who cares?
- 3 Perspectives
- Examples



What are
NGBs and
why should
we care?

=> Motivation, basics, examples, critical properties,...

NGB Electrolytes – from fundamentals to application

- General functionalities
- Special NGB demands & limits
- Common measures
- Examples



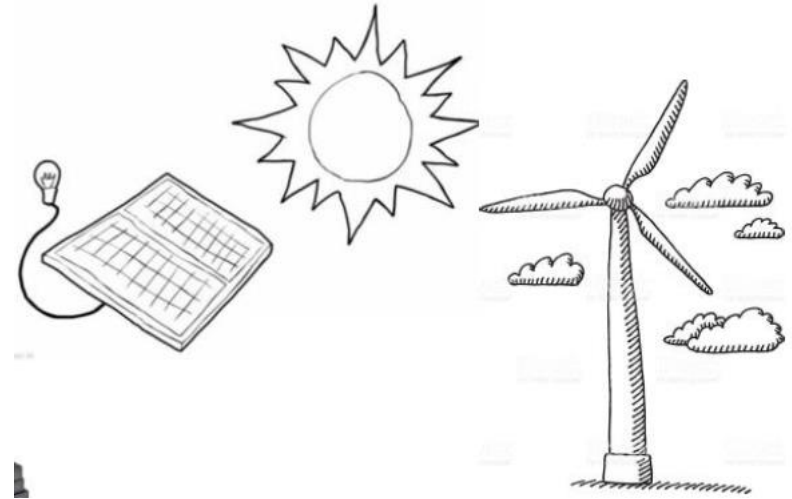
How are NGB
electrolytes
composed
and why?

=> Material choices and motivations, differences, problems,...





Top 10 Emerging Technologies of 2016

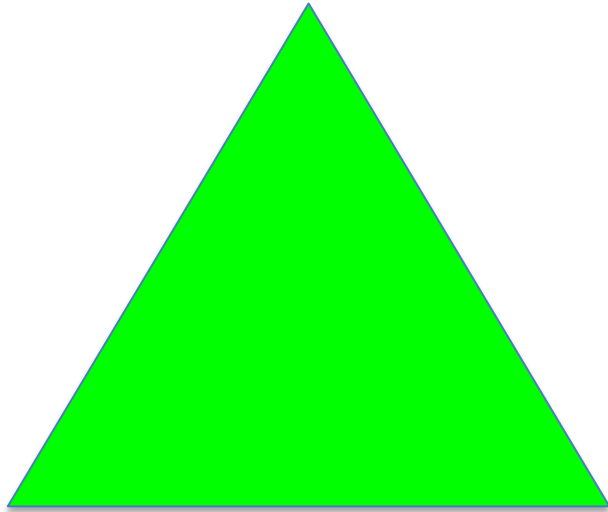


1. Nanosensors and the Internet of Nanothings – With the Internet of Things expected to comprise 30 billion connected devices by 2020, one of the most exciting areas of focus today is now on nanosensors capable of circulating in the human body or being embedded in construction materials. Once connected, this Internet of Nanothings could have a huge impact on the future of medicine, architecture, agriculture and drug manufacture.



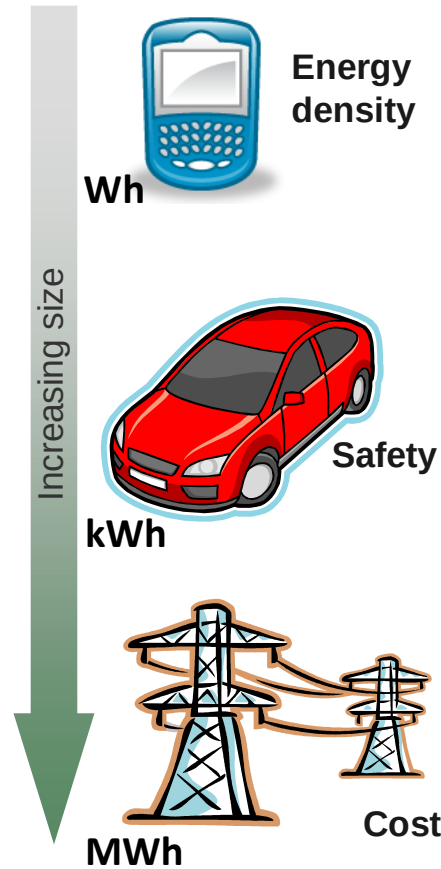
2. Next Generation Batteries – One of the greatest obstacles holding renewable energy back is matching supply with demand, but recent advances in energy storage using sodium, aluminium and zinc based batteries makes mini-grids feasible that can provide clean, reliable, round the clock energy sources to entire villages.



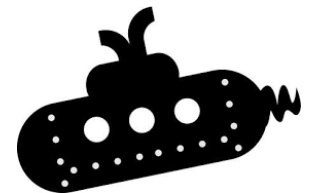


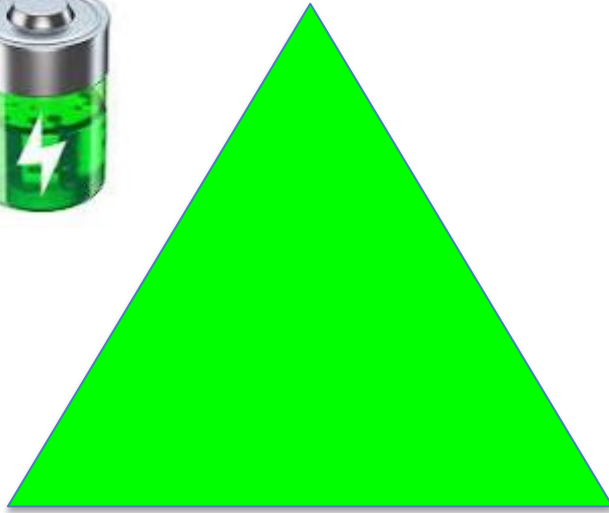
Performance

Energy
Power
Life-length
Safety
Cost
...
Driving range
Charging time



"New" Applications
=>
New demands





Sustainability

Resources

Production

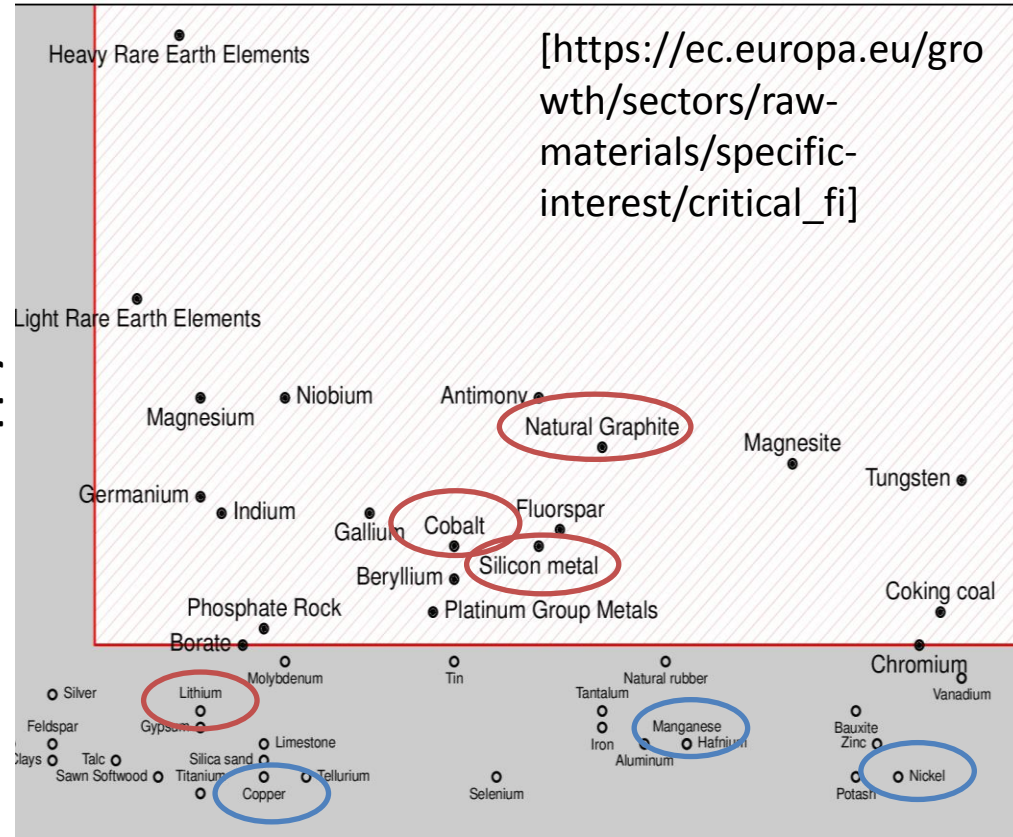
Efficiency

2nd use

Recycling

...

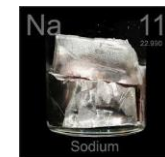
Supply Risk



Economic importance



20 ppm



22700 ppm

**Abundance
in Earth's crust:**





What if...?
How can...?
Can we...?
...

[illegible]

*Lanthanide series

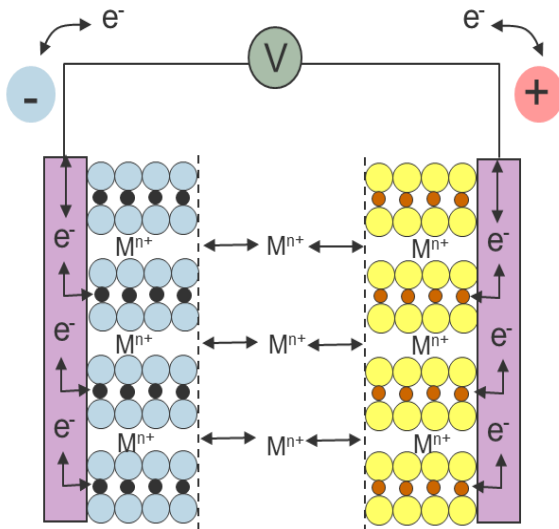
* * Actinide series

lanthanum 57	cerium 58	praseodymium 59	neodymium 60	promethium 61	samarium 62	europium 63	gadolinium 64	terbium 65	dysprosium 66	holmium 67	erbium 68	thulium 69	ytterbium 70
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb
138.91	140.12	140.91	144.24	145	150.36	151.96	157.25	158.93	162.50	164.93	167.26	168.93	173.04
actinium 89	thorium 90	protactinium 91	uranium 92	neptunium 93	plutonium 94	americium 95	curium 96	berkelium 97	californium 98	einsteinium 99	fermium 100	nobelium 101	lawrencium 102
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No
227	232.04	231.04	238.03	237.05	244.06	243.06	247.07	247.07	251.08	252.08	257	259	261





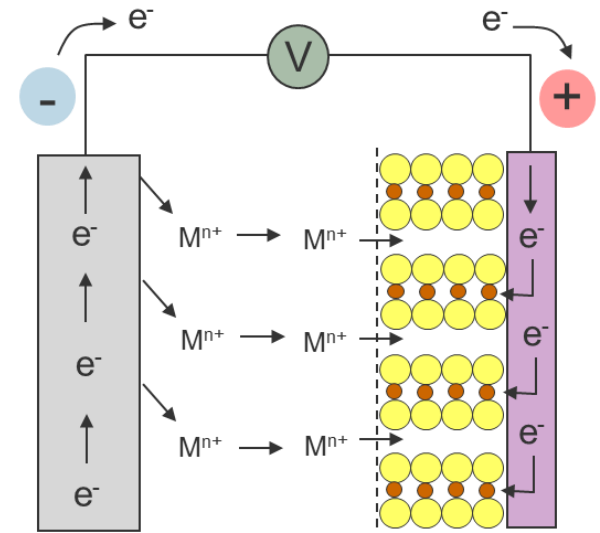
Metal-ion batteries: Na-ion (SIB/NIB)



Intercalation of M^+ into electrodes
No pure metal inside the battery (= LIBs)
Capacity not intrinsically = $f(\text{metal})$

Metal batteries:

- Li, Li-S, Li-O₂
- Na (Na-S)
- Mg, Ca, Al

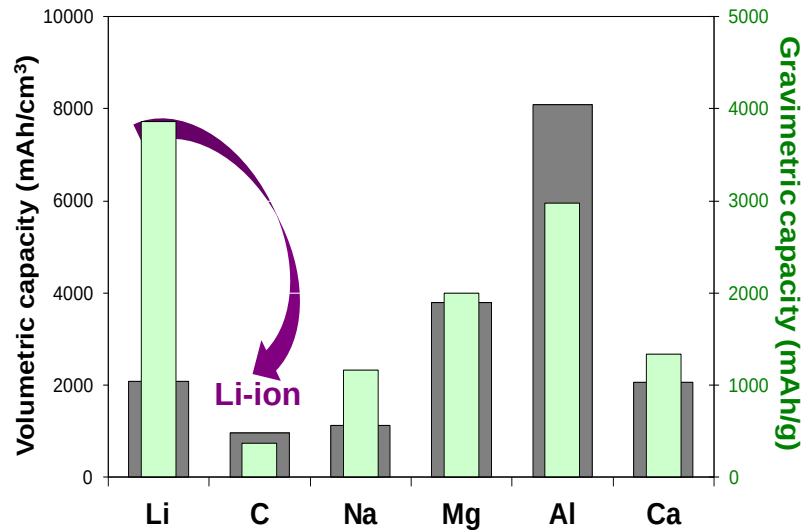
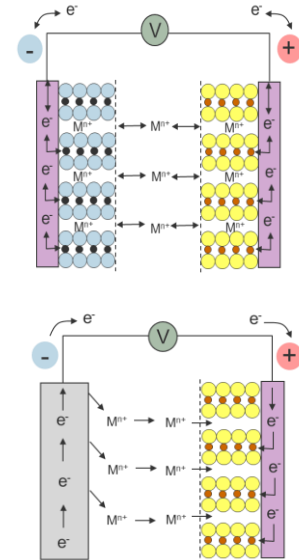
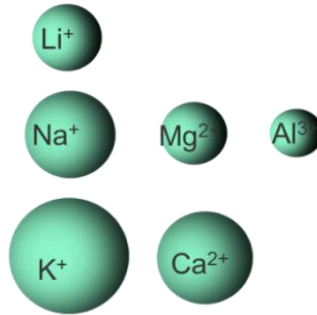


Stripping/plating of metal
Pure metal inside battery
Capacity intrinsically = $f(\text{metal})$





Energy $\approx$ Potential $\times$ Capacity



M Problems/Issues

- Li** Unsafe? TMO_x, S or O₂ cathodes?
- Na** Low melting/unsafe?! Na-S T_{op} $\approx$ 200°
- Mg** Stable electrolytes, migration
- Ca** Reversible plating/stripping, migration(?)
- Al** Reversible plating/stripping, migration

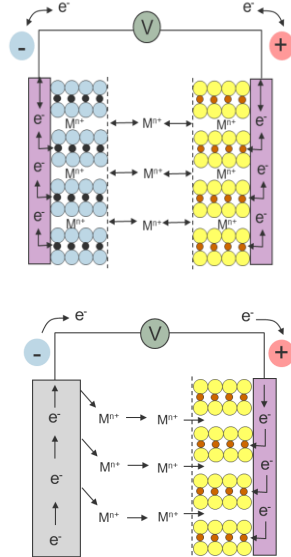
SIBs? Na in C host – as for LIB (graphite => HC)





=> **Electrolyte = Salt(s) + Solvent(s) + Additive(s)** <=

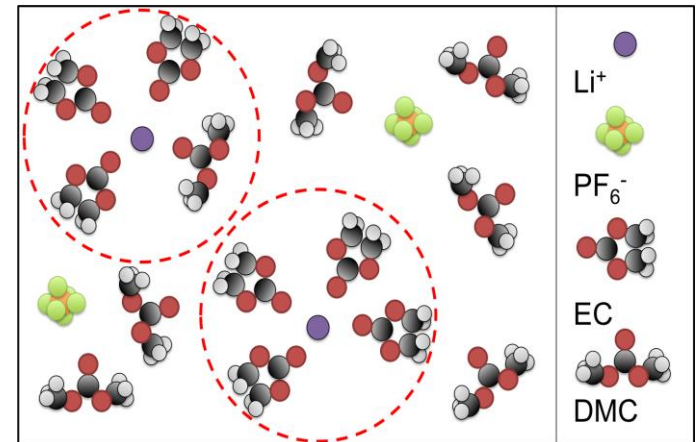
- **Mechanically stable – separate electrodes**
- **No e^- transport**
- **Rapid M^+ transport – ion conductivity**
- **Low viscosity – ion conductivity & cell-making**
- **Electrochemically (meta-)stable - ESW**
- **Acceptable T working range**



- **Sustainable**
- **Non-toxic**
- **Safe**
- **Low-cost**

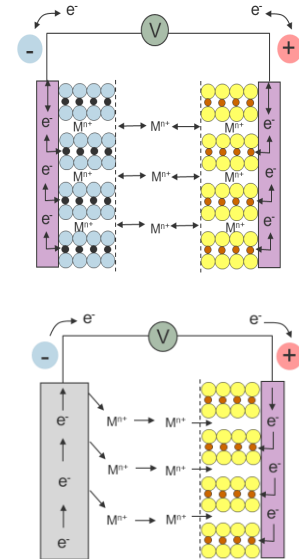
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Conventional LIB Electrolytes



“Standard”: 1 M LiPF_6 in EC:DMC



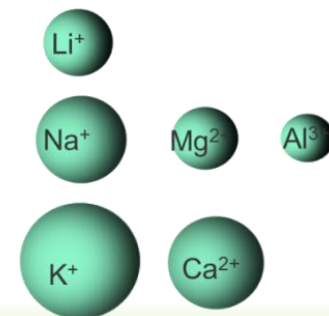
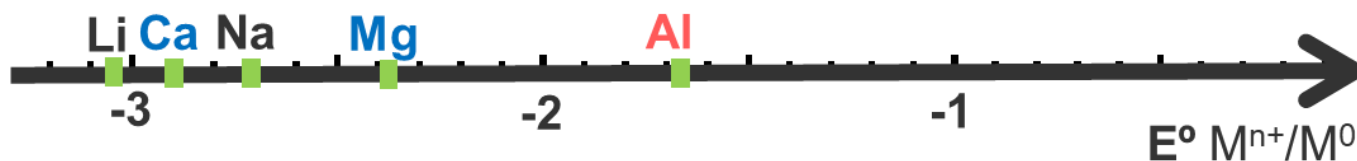
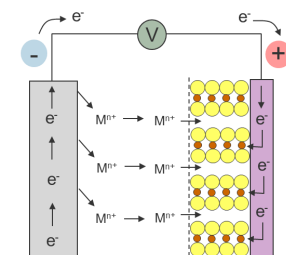
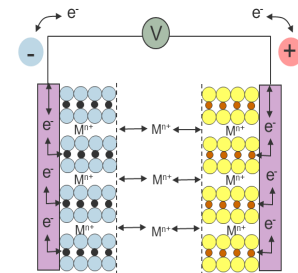


Discuss in groups of 3 persons for ca. 5 minutes

1. What do you know about electrolytes in general?
2. Do you think 1 M LiPF_6 in EC:DMC sounds OK?
3. What alternatives can you think of?



NGB	ESW	Solubility	Other
Li	≈LIB	≈LIB	Stable vs. Li (SEI)
Li-S	ca. 2 V	Not S, PS	Stable vs. Li (SEI)
Li-O ₂	ca. 3 V	Yes O ₂	Stable vs. Li ₂ O ₂ , Li (SEI)
Na-ion	≈LIB-0.3 V	≈LIB	
Na	≈LIB-0.3 V	≈LIB	Stable vs. Na, high T _{op} ?
Mg	ca. 3-4 V	salt 0.5 M	Stable vs. Mg
Ca	≈LIB(?)	salt 0.5 M	
Al	reduced, shifted	salt 0.5 M	[AlCl ₄] ⁻ transport?





Ion conductivity

“1 mS/cm” AC imp., dielectric spectroscopy

Thermal stability

“-30 - +50 °C mS/cm” TGA (iso-thermal), DSC

ESW

LSV, CV

Solubility

UV-VIS/cycling for S & PS, microelectrodes & DEMS for O₂

Density, Viscosity

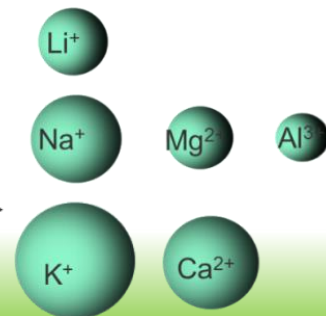
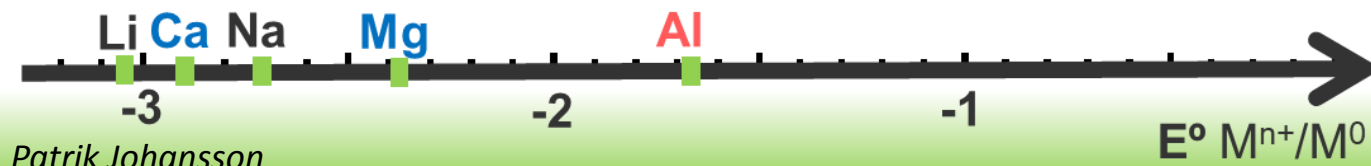
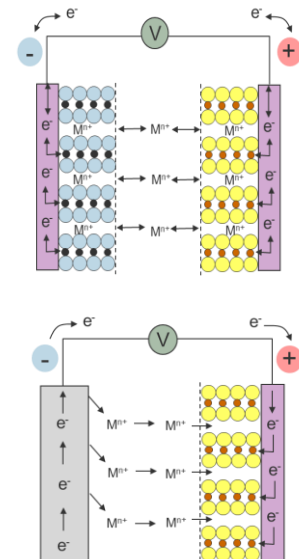
limitations in Wh/kg, transport

NMR, IR, Raman

ion transference/mobility, speciation

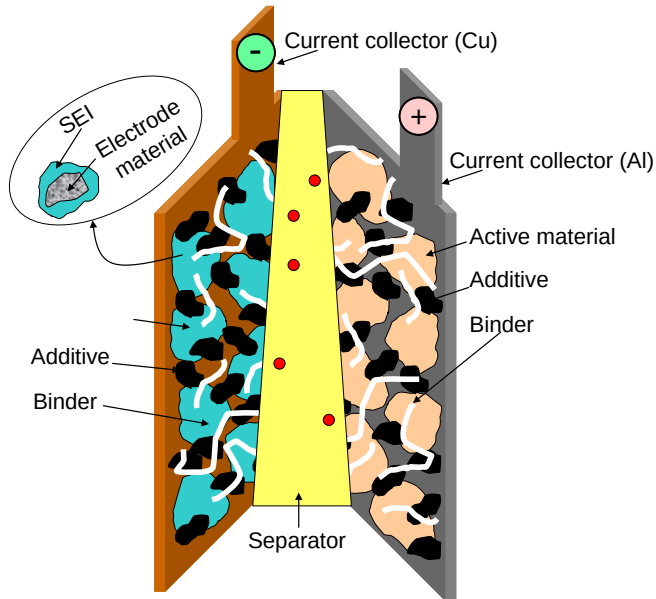
Cell cycling

...





LIB



Cu-graphite

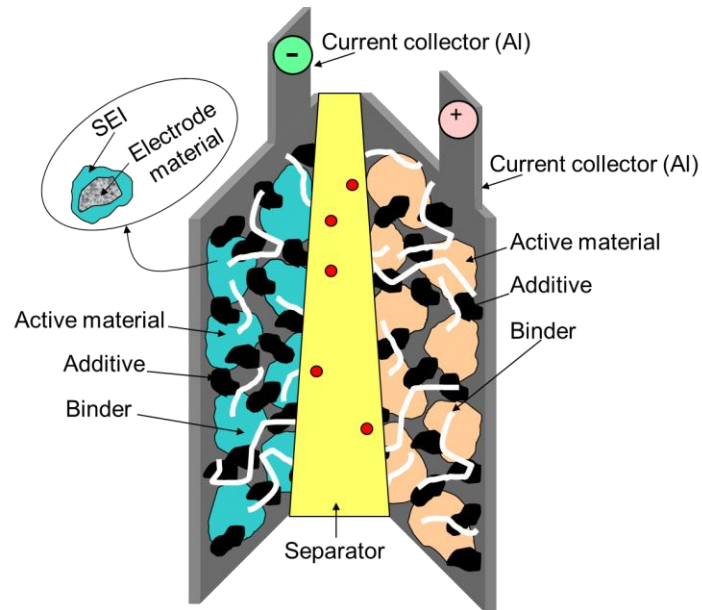
Same

1 M LiPF_6 in EC:DMC

Different

PC exfoliates graphite

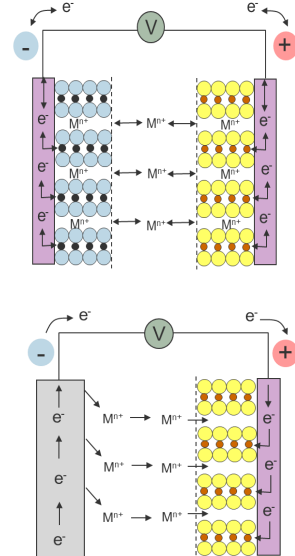
SIB



Al-HC

1 M NaPF_6 in EC:DMC

PC can be (and is) used



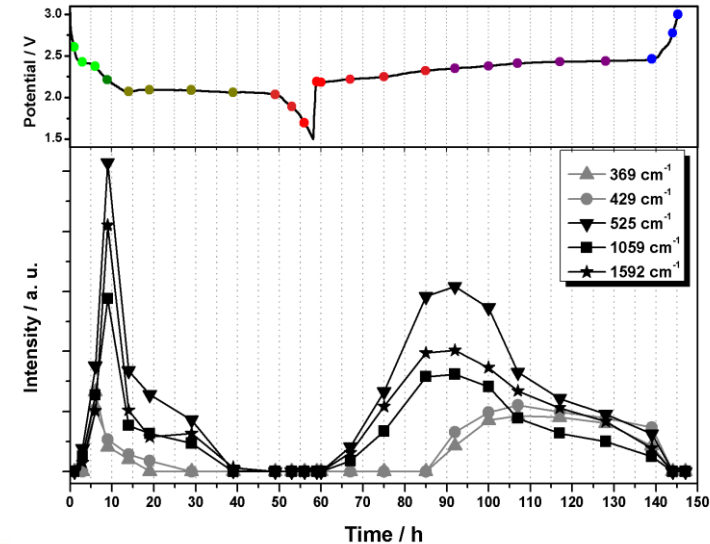
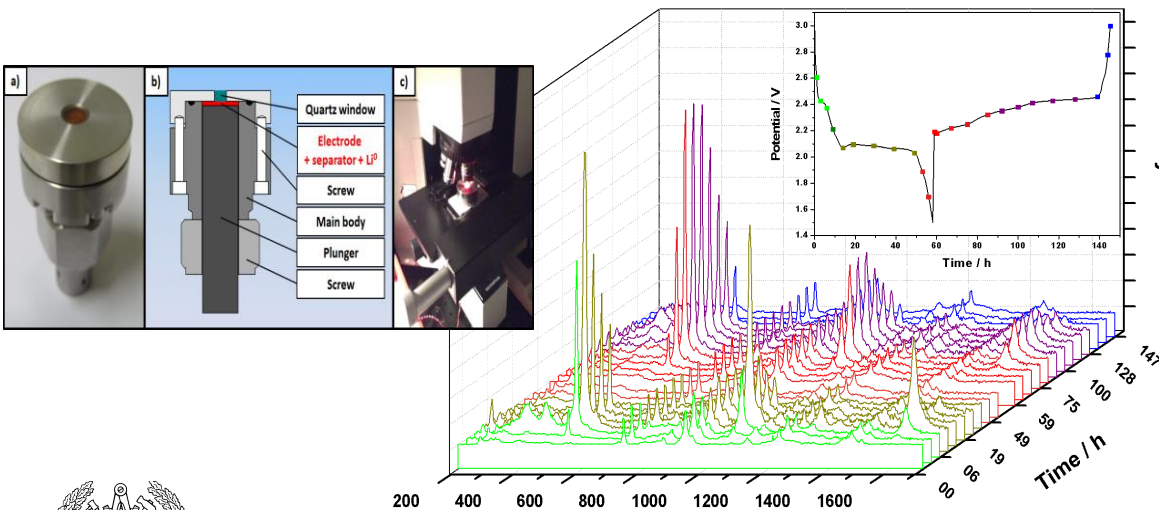
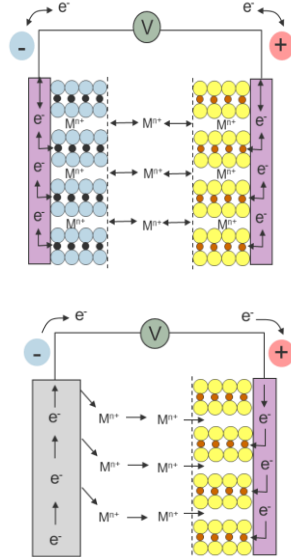
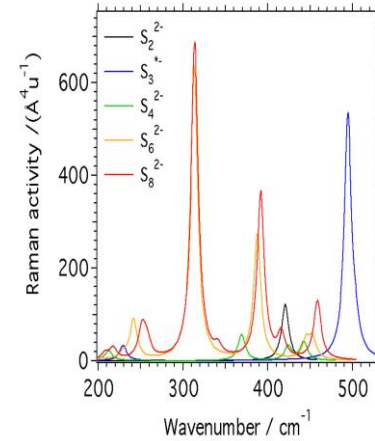
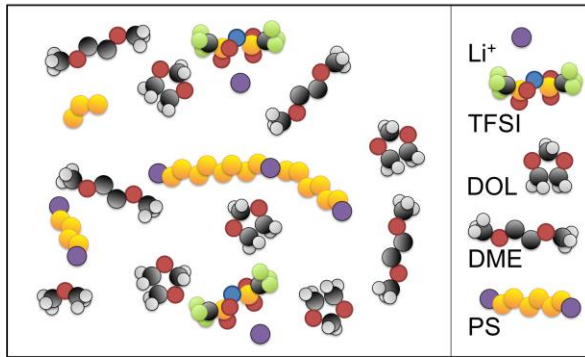


Solubility of S and PS = f(SOC) & cell composition

Cell: Li | 1 M LiTFSI in DOL:DME | C/S composite

Here: *Operando* confocal Raman + DFT comp.

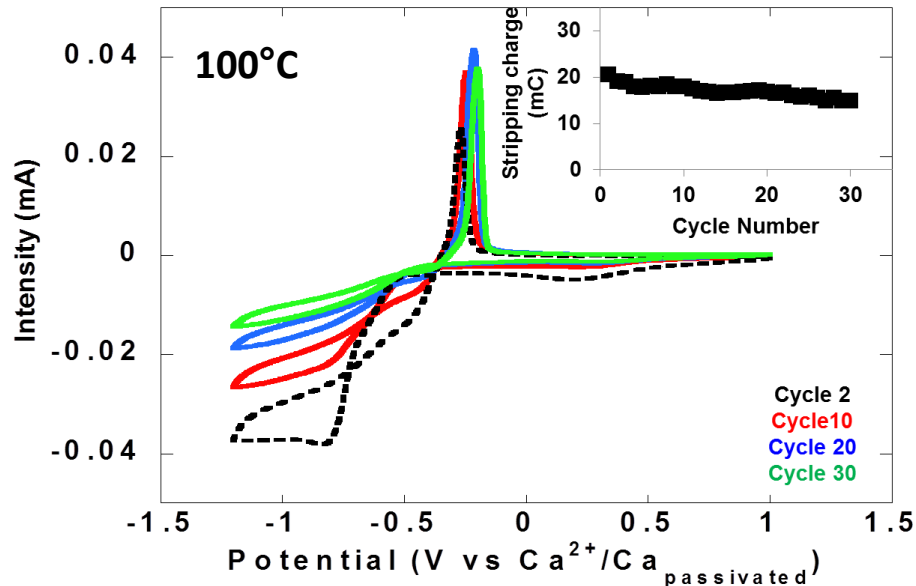
Li-S Battery Electrolytes





Stripping/plating of Ca difficult

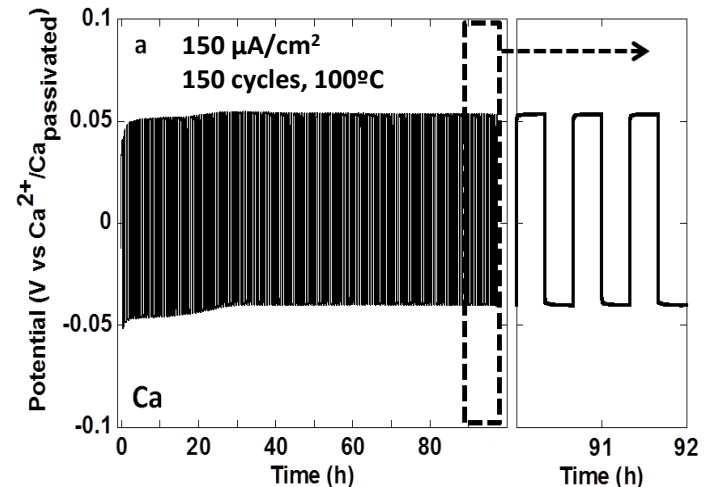
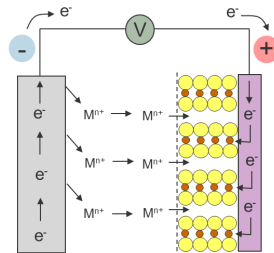
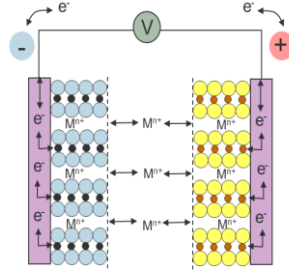
Half-cell: Ca | 0.45 M $\text{Ca}(\text{BF}_4)_2$ in EC:PC, $T_{\text{op}} = 100^\circ\text{C}$



☺ Capacity stable

☹ Large polarisation

☹ Electrolyte decomposes





NGB electrolytes:

- are often rather similar – also to LIB electrolytes
- have a few or more unique demands, esp. Li-O₂, Li-S
- stability is “always” an issue – esp. vs. metal electrode
- solubility is tricky to handle/measure *realistically*
- progress is in need for many tools and techniques

Concepts/Things I have not (yet) talked about:

- Ionic liquid based electrolytes
- Aqueous electrolytes
- Solid-state electrolytes (ceramics, glasses, polymers)
- ...





Thanks!

What questions do you have now?



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